A standalone proof of the lower endpoint in Proposition 7 of the Ringmin asymptotic sequel

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Abstract

We give a self-contained proof of the explicit lower bound $C_* \geq C_{\mathrm{term}} + \eta_{\mathrm{width}}$ for the asymptotic central-circle constant of Ringmin. Three nested terminal deletions of one cyclic order share one quadratic energy. An explicit dual potential, signed deletion estimate, crossing lemma, scalar minimax and global cubic certificate yield the bound. All constants and directed rational checks are specified here and in two small included arithmetic scripts. In particular,

$$0.14056946887766098063257 < C_{\mathrm{term}} + \eta_{\mathrm{width}} < 0.14056946887766098063392.$$

The coefficient is a lower bound, not an identified geometric optimum. This supplement reconstructs Proposition 7 of arXiv:2609.13630 and does not revise that manuscript or assert external scientific acceptance.

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1 Statement, notation and proof dependencies

The Ringmin problem places circles of radii $1, \dots, n$ externally tangent to a central circle of radius R, requiring the interiors of every pair of outer circles to be disjoint. Write $R^*(n)$ for the infimum of such feasible R. For a cyclic order ω of at least three distinct positive radii, $R_{\mathrm{chain}}(\omega)$ is the adjacent angular closure root defined in Section 2. It is only a necessary bound for a full feasible placement of that order.

Let τ be the unique solution of $\cos\tau = \tau$ in $(0,1)$ and put

$$q = \frac{1-\sin\tau}{1+\sin\tau}, \qquad C_{\mathrm{term}} = \frac{\tau(1+q)}{2\pi}, \qquad (\beta_1,\beta_2,\beta_3) = \left(\frac{2091}{10000}, \frac{10907}{50000}, \frac{23}{100}\right). \tag{1}$$

Define

$$D_i = \int_q^{\beta_i} \left[1+q-x-2\sqrt{x(1+q-x)}\right] \mathrm{d}x, \tag{2}$$

$$a = \beta_2 - \beta_1 = \frac{113}{12500}, \qquad b = \beta_3 - \beta_2 = \frac{593}{50000},$$

$$\mathcal{P} = \{h \in [0,\infty)^3 : h_1 + h_2 \le a,\ h_2 + h_3 \le b\},$$

$$F(h) = \sum_{i=1}^{3} h_i D_i - 8\sum_{i=1}^{3} h_i^3, \qquad \mathcal{L}(h) = 16 + 432\sum_{i=1}^{3} h_i, \tag{3}$$

$$\eta(h) = \frac{(F(h))_+}{\pi\mathcal{L}(h)}, \qquad \eta_{\mathrm{width}} = \max_{h\in\mathcal{P}} \eta(h), \quad r_+ = \max\{r, 0\}.$$

Theorem 1.1. *The limit $C_* = \lim_{n\to\infty} R^*(n)/n^2$ exists. The maximum in* (3) *is positive and uniquely attained at $u = (a - z_*, z_*, b - z_*)$, where z_* is the unique root on $[0, a]$ of the polynomial G defined in* (21)*. Moreover*

$$C_* \ge C_{\mathrm{term}} + \eta_{\mathrm{width}},$$
$$\frac{1405694688776609806 3257}{10^{23}} < C_{\mathrm{term}} + \eta_{\mathrm{width}} < \frac{1405694688776609806 3392}{10^{23}}. \tag{4}$$

Every terminating decimal in this document denotes an exact rational. We use $\alpha = k/n$ for the finite grid's lower endpoint; a, b are reserved for the two width constraints. The root z_* belongs solely to this lower bound and has no connection to an upper-bound parameter. The proof is organized as follows.

Input	Complete argument	Dependency
L1	Angular geometry and deletion (§2)	definitions
L2	Exact terminal parameters (§3)	elementary calculus
L3	Dual and common energy (§4)	L2 grid domain
L4	Signed deletion and floors (§5)	L3
L5	Shared crossings (§6)	common measure
L6	Finite minimax (§7)	L1, L3–L5
L7	Width optimum and enclosure (§8, §9)	L2
L8	Ordered limits (§10)	L6, L7; limit existence

The final limit-existence argument is included in Section 11. The references identify provenance; no external proof or operational log is needed to follow the arguments below.

2 Angular geometry and deletion (L1)

For radii $u, v > 0$ the angle required by non-overlap is

$$\theta_R(u,v) = 2\arcsin\frac{\sqrt{uv}}{\sqrt{(R+u)(R+v)}} = 2\arctan\frac{\sqrt{uv}}{\sqrt{R(R+u+v)}}. \tag{5}$$

Indeed the centers lie at distances $R+u, R+v$ from the origin. The cosine law says that their distance is at least $u+v$ exactly when the smaller angular separation is at least (5). The second identity uses $1-uv/((R+u)(R+v)) = R(R+u+v)/((R+u)(R+v))$. Both directed angular arcs must therefore be at least $\theta_R(u,v)$.

Lemma 2.1 (Uniform root sandwich)**.** *For a cycle ω of at least three radii in $\{1,\dots,n\}$, put $S(\omega) = \sum_{\{u,v\}\in\omega}\sqrt{uv}$, counting every cyclic edge once, including the closing edge. Then the equation $\sum_\omega \theta_R(u,v) = 2\pi$ has a unique positive root and*

$$\frac{S(\omega)}{\pi} - n \le R_{\mathrm{chain}}(\omega) \le \frac{S(\omega)}{\pi}. \tag{6}$$

Proof. Each angle is continuous and strictly decreasing from π to zero as R increases from zero to infinity. The sum starts above 2π. For $1 \le u, v \le n$, $\arcsin t \ge t$, $\arctan t \le t$ and (5) give

$$\frac{2\sqrt{uv}}{R+n} \le \theta_R(u,v) \le \frac{2\sqrt{uv}}{R}.$$

At the root these give $S/(R+n) \le \pi \le S/R$, which is precisely (6). No scale assumption or parity restriction is used. □

Lemma 2.2 (Deletion from full geometry)**.** *Delete any outer circles from a full feasible configuration, retaining at least three circles with their original radii. If ω' is the cyclic order of the survivors, then $R_{\mathrm{chain}}(\omega') \le R$.*

Proof. Every surviving pair remains non-overlapping. If g is the directed gap between successive surviving centers, then $\min\{g, 2\pi - g\} \ge \theta_R(u,v)$, hence $g \ge \theta_R(u,v)$. All these gaps, including closure, sum to 2π. Thus the chain sum at R is at most 2π; strict decrease gives the assertion. This proof uses full pair feasibility before deletion. It never reconstructs a full placement from a chain root. □

3 Exact terminal parameters (L2)

Lemma 3.1. *The parameters in (1) are uniquely defined, with $3/17 < q < 1/5 < \beta_1$. Furthermore*

$$J := \int_q^1 \sqrt{x(1+q-x)}\,\mathrm{d}x = \pi C_{\mathrm{term}}. \tag{7}$$

Among $0 < r < 1$ the expression $\pi^{-1}\int_r^1 \sqrt{x(1+r-x)}\,\mathrm{d}x$ has a unique global maximum at $r = q$.

Proof. The function $\cos t - t$ is continuous, positive at zero, negative at one and strictly decreasing on $(0,1)$, since its derivative is $-1-\sin t$. This proves existence and uniqueness of τ. The rational gates in Section 9 prove $3/17 < q < 1/5$ without assuming rounded approximations.

For arbitrary $r \in (0,1)$ put $v = (1-r)/(1+r)$. Substitution $y = (2x-1-r)/(1+r)$ gives

$$\frac{1}{\pi}\int_r^1 \sqrt{x(1+r-x)}\,\mathrm{d}x = \frac{\arcsin v + v\sqrt{1-v^2}}{\pi(1+v)^2} =: \Phi(v).$$

Here $r \mapsto v$ is a bijection of $(0,1)$ onto itself, reversing order. Writing $A(v) = \arcsin v + v\sqrt{1-v^2}$ gives

$$A'(v) = 2\sqrt{1-v^2}, \qquad \Phi'(v) = \frac{2(\sqrt{1-v^2} - \arcsin v)}{\pi(1+v)^3}.$$

With $t = \arcsin v$, the numerator changes sign exactly at $t = \tau$. Thus $v = \sin\tau$ is the unique global maximum. The endpoint limits are 0 and $1/8$, and strict monotonicity on either side excludes boundary maximizers. At this point $\sqrt{1-v^2} = \tau$, so $A(v) = \tau(1+v)$ and $\Phi(v) = \tau/(\pi(1+v))$. Since $1+q = 2/(1+v)$, this proves (7) and the claimed value. Each D_i is now an ordinary integral of a continuous, exactly specified function over a uniquely specified interval. Its sign and enclosure are proved arithmetically in Section 9. □

The optimization just made concerns a terminal integral. The lower-bound proof below uses its identity (7); no theorem about an exactly minimizing finite Supnick cycle is required as an additional premise.

4 One outer order, its dual and its energy (L3)

Let $n \geq 102$, $k = \lfloor qn \rfloor$, $\alpha = k/n$, $s = 1+\alpha$, $V = \{k/n, (k+1)/n, \ldots, 1\}$. The strict bound $q > 3/17$ gives

$$\alpha > q - 1/n > 3/17 - 1/102 = 1/6, \qquad \alpha \leq q < 1/5. \tag{8}$$

Fix one genuine cyclic order σ of the original labels $k, \ldots, n$. Use normalized coordinates $x = j/n$ in its edges and set

$$f(x,y) = \sqrt{xy}, \qquad r(x) = s - x,$$
$$W_0 = \frac{1}{n}\sum_{\{x,y\}\in\sigma} f(x,y), \quad J_n = \frac{1}{n}\sum_{x\in V} f(x, r(x)), \quad e = W_0 - J_n.$$

Thus $W_0 = S(\sigma)/n^2$. The reflection r permutes V exactly.

Lemma 4.1 (Assignment dual)**.** *For $c = s/2$ define*

$$\psi(x) = \frac{c}{2} + \frac{1}{2}\int_c^x \sqrt{\frac{s-t}{t}}\,\mathrm{d}t.$$

Then for $x, y \in [\alpha, 1]$,

$$\frac{(x+y-s)^2}{8} \leq f(x,y) - \psi(x) - \psi(y) \leq \frac{(x+y-s)^2}{8\alpha}. \tag{9}$$

Consequently

$$e \geq 0, \qquad E := \frac{1}{n}\sum_{\{x,y\}\in\sigma} (x+y-s)^2 \leq 8e. \tag{10}$$

Proof. Differentiation shows that $\psi(x) + \psi(s-x)$ and $f(x, s-x)$ have equal derivatives; at $x = c$ both equal c. For $H(x,y) = f(x,y) - \psi(x) - \psi(y)$ this gives $H(x, s-x) = 0$, and

$$\partial_y H(x,y) = \frac{x+y-s}{2\sqrt{y}(\sqrt{x} + \sqrt{s-y})}.$$

The denominator lies between 4α and 4. Integrate from $s-x$ to y. If $y < s-x$, the integration direction and numerator both reverse sign. In both cases integrating the linear numerator gives (9). Reflection and degree two imply $\sum_{x\in V} f(x, r(x)) = 2\sum_{x\in V}\psi(x) = \sum_{\{x,y\}\in\sigma}(\psi(x) + \psi(y))$. Summing (9) proves (10). □

The reference J_n need not be the cost of a realizable cycle: reflection can have two-cycles and a middle fixed point. Only its exact assignment identity was used. No loop or duplicate neighbor is inserted into σ.

Replace each cyclic edge by its two orientations and define

$$\mu = \frac{1}{2n} \sum_{(x,y) \text{ oriented edge}} \delta_{(x,r(y))}, \qquad \nu = \frac{1}{n} \sum_{x \in V} \delta_x. \tag{11}$$

Both marginals of μ are ν, since each vertex occurs twice in each endpoint position and reflection permutes the grid. Hence

$$\int [A(x) - A(z)] \, \mathrm{d}\mu(x,z) = 0, \qquad \int (x-z)^2 \, \mathrm{d}\mu(x,z) = E \tag{12}$$

for any function A on V. The total mass is $(n-k+1)/n$, not artificially renormalized to one. Every restriction below uses this same σ, μ and E.

5 Signed deletion, seams and finite errors (L4)

Consider any $\beta \in (q, 23/100]$ and let $\ell = \lfloor \beta n \rfloor$. Assume in addition

$$\ell/n > q, \qquad \ell \le n-2. \tag{13}$$

These gates ensure a nonempty deleted set and at least three survivors. Write $\gamma = \ell/n$ and $B = (\ell - 1/2)/n$. A grid vertex is deleted exactly when $x < B$. Since $s - 2\gamma > 7/6 - 46/100 = 53/75 > 1/2$, deleted vertices lie below $s/2$. Put

$$D_\alpha(x) = s - x - 2\sqrt{x(s-x)}, \qquad D_n = \frac{1}{n} \sum_{j=k}^{\ell-1} D_\alpha(j/n),$$

$$W_\beta = \frac{1}{n} \sum_{\{x,y\} \in \sigma | \{\ell,\dots,n\}} f(x,y), \qquad \Delta = W_\beta - W_0 - D_n,$$

$$K = \int |x - z| \mathbf{1}_{(x-B)(z-B)<0} \, \mathrm{d}\mu(x,z).$$

Lemma 5.1 (Midpoint strip mass)**.** *For every $h > 0$, $\nu\{x : |x - B| \le h\} \le 4h$.*

Proof. No grid point is within $1/(2n)$ of B. Below this radius the set is empty. Otherwise it has at most $2nh + 1$ points, of total mass at most $2h + 1/n \le 4h$. Truncating the grid at either endpoint only reduces mass. □

Lemma 5.2 (Signed first variation)**.** *Let*

$$p(x) = \tfrac{1}{2}\left(1 - \sqrt{x/(s-x)}\right), \quad P(x) = p(x)\mathbf{1}_{x<B}, \quad \Lambda = \frac{1}{n} \sum_{x<B} p(x) \sum_{y \sim x} (x + y - s).$$

The neighbors $y \sim x$ are those in the original outer cycle. Then

$$\Lambda = 2 \int P(x)(x - z) \, \mathrm{d}\mu, \qquad |\Delta - \Lambda| \le 43E. \tag{14}$$

Proof. The first identity follows directly from the oriented mass $1/(2n)$. Partition deleted vertices into maximal cyclic runs. Rotate the written cycle to a survivor if necessary. Every removed edge and every replacement edge is then accounted for exactly once, including the seam.

For an isolated deleted vertex x with surviving neighbors y, z the change in unnormalized f-cost is $g(x; y, z) = f(y, z) - f(x, y) - f(x, z)$. At $y = z = t = s - x$ it equals $D_\alpha(x)$, and both first derivatives in y, z equal $p(x)$. The Hessian entries are

$$g_{yy} = \frac{\sqrt{x} - \sqrt{z}}{4y^{3/2}}, \quad g_{zz} = \frac{\sqrt{x} - \sqrt{y}}{4z^{3/2}}, \quad g_{yz} = \frac{1}{4\sqrt{yz}}.$$

On $[\alpha, 1]^3$ their absolute values are respectively less than $4, 4, 2$: $1/(4\alpha^{3/2}) < 3\sqrt{6}/2 < 4$ and $1/(4\alpha) < 3/2 < 2$. The symmetric Hessian therefore has operator norm at most 6 (each absolute row sum is at most 6). Taylor's integral remainder along the segment from (t, t) to (y, z) yields

$$|g - D_\alpha(x) - p(x)(y - t + z - t)| \le 3((y - t)^2 + (z - t)^2).$$

The segment stays in $[\alpha, 1]^2$. Across isolated deletions each incident edge occurs at most once, so their total normalized error is at most $3E$.

For runs of length $m \ge 2$, let M_{LL} be the total number of outer edges with both endpoints deleted and V_{bad} the number of vertices in these runs. Since $m \le 2(m - 1)$, $V_{\text{bad}} \le 2M_{LL}$. A run replaces $m + 1$ edges by one edge; all weights lie in $[0, 1]$, so its cost change has absolute value at most $m + 1 \le 2m$. Since $|D_\alpha(x)| \le 2$, its discrepancy from the m ideal changes is at most $4m$. The total normalized discrepancy of these runs is at most $8M_{LL}/n$. For a deleted vertex, $0 < p(x) < 1/2$ and each defect has absolute value at most $1 - \alpha < 1$. The part of Λ from bad vertices therefore has absolute value at most $V_{\text{bad}}/n \le 2M_{LL}/n$; this includes both incidences of each LL edge. On each such edge $|x + y - s| > 1/2$, so $M_{LL}/n \le 4E$. Combining contributions gives $3E + (8 + 2)M_{LL}/n \le 43E$. □

Lemma 5.3 (Cancellation and signed deletion)**.** *Under the same hypotheses,*

$$|\Lambda| \le 11E + 2K, \qquad |W_\beta - W_0 - D_n| \le 54E + 2K. \tag{15}$$

Proof. Throughout $[\alpha, 1]$, $|p(x)| < 1$ and

$$p'(x) = -\frac{s}{4\sqrt{x}(s - x)^{3/2}}, \qquad |p'(x)| \le \frac{s}{4\alpha^2} < \frac{54}{5} < 11.$$

The first bound follows from $\sqrt{x/(s - x)} < \sqrt{6} < 3$; the second uses $s < 6/5$ and $\alpha > 1/6$. Let $A(u) = \int_\alpha^u P(v)\,\mathrm{d}v$, a continuous primitive with a piecewise continuous derivative. For grid points x, z, integrate

$$P(x) - P(v) = \mathbf{1}_{x<B}(p(x) - p(v)) + p(v)(\mathbf{1}_{x<B} - \mathbf{1}_{v<B})$$

along the segment from z to x. The smooth term contributes at most $11|x - z|^2/2$. The jump term vanishes unless the segment crosses B and otherwise contributes at most $|x - z|$. Both orientations give

$$|P(x)(x - z) - (A(x) - A(z))| \le \tfrac{11}{2}(x - z)^2 + |x - z|\mathbf{1}_{(x-B)(z-B)<0}.$$

Neither endpoint equals the midpoint B. Equal marginals in (12) cancel the primitive exactly. Multiply the remaining bound by two and use (14) to obtain $11E + 2K$, then add $43E$. Thus the constant is $43 + 11 = 54$, with no mesh error in this estimate. □

Lemma 5.4 (Uniform floor estimates)**.** *Set $D(\beta) = \int_q^\beta D_q(x)\,\mathrm{d}x$. For every cutoff satisfying* (13)*,*

$$|J_n - \pi C_{\text{term}}| \le 5/n, \qquad |D_n - D(\beta)| \le 19/(2n). \tag{16}$$

Proof. Let $F_\alpha(x) = \sqrt{x(1+\alpha-x)}$. All factors in the comparisons below lie in $[1/6, 1]$, and direct differentiation gives

$$|\partial_x F_\alpha| \le 3, \quad |\partial_\alpha F_\alpha| \le 3/2, \quad |F_\alpha| \le 1, \qquad |\partial_x D_\alpha| \le 7, \quad |\partial_\alpha D_\alpha| \le 2, \quad |D_\alpha| \le 2.$$

For example each coordinate derivative of $\sqrt{xt}$ is at most $\sqrt{6}/2 < 3/2$, and $\partial_\alpha D_\alpha = 1 - \sqrt{x/(1+\alpha-x)}$. For a Lipschitz function with constant M, the left-rectangle error on each mesh interval is at most $M/(2n^2)$. There are at most n intervals. The inclusive sum J_n also has its last endpoint term; its error against $\int_\alpha^1 F_\alpha$ is at most $(3/2+1)/n$. Removing $[\alpha, q]$ costs at most $1/n$ and changing the parameter from α to q on $[q, 1]$ costs at most $3/(2n)$. Their sum is $5/n$.

The sum D_n is exactly a left sum on $[\alpha, \gamma]$; its error is at most $7/(2n)$. Removing $[\alpha, q]$, changing the parameter on $[q, \gamma]$ and adding $[\gamma, \beta]$ cost at most $2/n$ each. Here $\gamma > q$ is an explicit hypothesis. Their sum is $19/(2n)$. Only $0 \le q - \alpha < 1/n$ and $0 \le \beta - \gamma < 1/n$ are used. No numerical evaluation of a transcendental floor or parity selection occurs. □

6 One shared crossing budget (L5)

Lemma 6.1. *Let $B_1 < \cdots < B_m$, $h_i > 0$, and let μ be a finite nonnegative measure with first marginal ν, finite energy $E = \int |x-z|^2 \,\mathrm{d}\mu$ and strip bounds $\nu\{|x - B_i| \le h_i\} \le 4h_i$. Define $K_i = \int |x-z| \mathbf{1}_{(x-B_i)(z-B_i)<0} \,\mathrm{d}\mu$. If $h_i + h_{i+1} \le B_{i+1} - B_i$ for every adjacent pair, then*

$$\sum_{i=1}^{m} h_i K_i \le 4 \sum_{i=1}^{m} h_i^3 + E. \tag{17}$$

Proof. Write $d = |x-z|$. At cutoff i split crossings into short ($d \le h_i$) and long ($d > h_i$). A short crossing has $|x - B_i| \le h_i$, so its weighted integrand is at most $h_i^2 \mathbf{1}_{|x-B_i|\le h_i}$. Integrating and summing gives at most $4\sum h_i^3$. The strictly crossed cutoffs form a consecutive block $p, \ldots, r$. With no crossing the long contribution is zero; with one it is at most $h_p d < d^2$ if long. If $r > p$, summing adjacent separation gives

$$\sum_{i=p}^{r} h_i = \sum_{i=p}^{r-1} (h_i + h_{i+1}) - \sum_{i=p+1}^{r-1} h_i \le B_r - B_p - \sum_{i=p+1}^{r-1} h_i \le B_r - B_p < d.$$

Thus every crossing in this block is long, and their combined weighted integrand is $d\sum_{i=p}^{r} h_i \le d^2$. Integration spends E once, even when the same pair crosses all three thresholds. The equality case $d = h_i$ was included in the short part; endpoints at a cutoff do not strictly cross it. No pair is omitted or charged to several energy budgets. □

7 Finite minimax and transfer to the global problem (L6)

Use the fixed three cutoffs of (1). Let $\ell_i = \lfloor \beta_i n \rfloor$, $B_i = (\ell_i - 1/2)/n$ and $\sigma_i = \sigma|\{\ell_i, \ldots, n\}$, with $\sigma_0 = \sigma$. Put $W_i = S(\sigma_i)/n^2$ and let $D_{i,n}$ denote D_n at cutoff i. Define the common-order minimax

$$M_{n,4} = \min_{\sigma \text{ cyclic on } \{\lfloor qn \rfloor, \ldots, n\}} \max_{0 \le i \le 3} R_{\mathrm{chain}}(\sigma_i).$$

Only sufficiently large n are needed; all its cycles then have at least three positive radii. For positive widths define

$$H = \sum_i h_i, \quad Q_3 = \sum_i h_i^3, \quad L = 16 + 432H, \quad F_n = \sum_i h_i D_{i,n} - 8Q_3.$$

Proposition 7.1. *Suppose*

$$n \ge 102, \quad \ell_1/n > q, \quad \ell_1 < \ell_2 < \ell_3 \le n-2, \quad h_i + h_{i+1} \le \frac{\ell_{i+1} - \ell_i}{n} \quad (i = 1, 2). \tag{18}$$

Then

$$R^*(n) \ge M_{n,4}, \tag{19}$$

$$\frac{M_{n,4}}{n^2} \ge \frac{J_n + (F_n)_+/L}{\pi} - \frac{1}{n} \ge C_{\text{term}} + \eta(h) - \frac{A(h)}{n}, \tag{20}$$

$$A(h) = 1 + \frac{5 + 19H/(2L)}{\pi}.$$

Proof. All three cutoffs satisfy Section 5's hypotheses. With the same outer measure, Lemmas 5.3 and 6.1 give

$$\begin{aligned} \sum_i h_i W_i &\ge H(J_n + e) + \sum_i h_i D_{i,n} - 54HE - 2\sum_i h_i K_i \\ &\ge H(J_n + e) + F_n - (54H + 2)E \\ &\ge H(J_n + e) + F_n - Le. \end{aligned}$$

Here $L = 8(54H+2)$ because $E \le 8e$; the factor 16 is not multiplied by the number of cutoffs. Retaining the outer cost and the positive weighted mean of the inner costs yields

$$\max_{0 \le i \le 3} W_i \ge J_n + \max\left\{e, \frac{F_n}{H} - \left(\frac{L}{H} - 1\right)e\right\}.$$

Since $L > H > 0$, if $F_n > 0$ the increasing and decreasing branches meet at $e = F_n/L \ge 0$. Below that point the decreasing branch is at least its meeting value, and above it the increasing branch is at least that value. If $F_n \le 0$, the first branch is always nonnegative and $e = 0$ makes the maximum zero. The infimum over all $e \ge 0$ is therefore $(F_n)_+/L$ in every case. This scalar infimum need not be realized by a tour. The lower sandwich (6) and a finite minimum over orders prove the first inequality in (20).

Lemma 5.4 gives $F_n \ge F(h) - 19H/(2n)$ and $J_n \ge \pi C_{\text{term}} - 5/n$. The positive-part function is increasing and 1-Lipschitz, so $(F_n)_+ \ge F(h)_+ - 19H/(2n)$. This proves the second inequality even when F_n or $F(h)$ is nonpositive.

For (19), take any full feasible original placement at radius R. Delete the labels below $\lfloor qn \rfloor$ to obtain one common order σ. Its three further restrictions are exactly σ_i. Lemma 2.2 bounds each chain root by R. Thus $M_{n,4} \le R$ for every full feasible placement. Taking the infimum over R proves the assertion without an attainment assumption. □

All estimates are uniform in the order, include the seam and consecutive deleted runs, and apply to both cardinality parities. No implication from chain feasibility to full feasibility was used.

8 The unique global width optimum (L7)

Set $d = a + b$ and $\ell = 16 + 432d$; these are width constants, not grid parameters. On the edge where both separation constraints are active set

$$\begin{aligned} N(z) &= F(a - z, z, b - z) = c_0 + c_1 z - 24dz^2 + 8z^3, \\ c_0 &= aD_1 + bD_3 - 8(a^3 + b^3), \qquad c_1 = D_2 - D_1 - D_3 + 24(a^2 + b^2), \\ L(z) &= \ell - 432z, \\ G(z) &= N'(z)L(z) + 432N(z) \\ &= \ell c_1 + 432c_0 - 48\ell dz + (24\ell + 10368d)z^2 - 6912z^3. \end{aligned} \tag{21}$$

Its exact derivative is

$$G'(z) = N''(z)L(z) = 48(z-d)L(z) < 0 \qquad (0 \le z \le a), \tag{22}$$

since $a < d$ and $L(z) \ge 16 + 432b > 0$.

The rational computations proved in Section 9 give $G(0) > 0$, $G(a) < 0$ and the more precise opposite signs at

$$z_- = 0.00451910758124826, \qquad z_+ = 0.00451910758124827. \tag{23}$$

Together with (22), continuity proves the existence and uniqueness of $z_* \in (z_-, z_+)$. All coordinates of $u = (a - z_*, z_*, b - z_*)$ are strictly positive.

Lemma 8.1 (Global cubic certificate)**.** *Let $\rho = F(u)/\mathcal{L}(u)$ and $g_i = D_i - 24u_i^2 - 432\rho$. Then $F(u) > 0$ and*

$$\lambda_1 = g_1 > 0.00035, \qquad \lambda_2 = g_3 > 0.00059, \qquad (g_1, g_2, g_3) = (\lambda_1, \lambda_1 + \lambda_2, \lambda_2).$$

For every $h \in \mathcal{P}$,

$$\begin{aligned} F(h) - \rho\mathcal{L}(h) = \lambda_1(h_1 + h_2 - a) + \lambda_2(h_2 + h_3 - b) \\ - 8\sum_{i=1}^{3}(h_i - u_i)^2(h_i + 2u_i) \le 0, \end{aligned} \tag{24}$$

with strict inequality unless $h = u$. Thus u is the unique global maximizer of η on the entire closed polyhedron.

Proof. The positivity gates are rational consequences of the integral and root intervals in Section 9. Since $G(z_*) = 0$, division by $L(z_*)$ gives $N'(z_*) + 432\rho = 0$. On substituting N' this is $-g_1 + g_2 - g_3 = 0$. The gradient of $F - \rho\mathcal{L}$ at u is therefore the stated positive combination of the two constraint normals. For each coordinate the exact identity

$$h_i^3 - u_i^3 - 3u_i^2(h_i - u_i) = (h_i - u_i)^2(h_i + 2u_i)$$

gives (24), since $F(u) - \rho\mathcal{L}(u) = 0$ and the two constraints are equalities at u. The linear terms are nonpositive on $\mathcal{P}$. Because $h_i \ge 0$ and $u_i > 0$, the cubic remainder is strictly negative if any coordinate differs. Hence $F(h)/\mathcal{L}(h) < \rho$ for $h \ne u$. If $F(h) \le 0$, the positive-part objective is zero and is also strictly smaller than $\rho/\pi > 0$. This covers zero widths, all faces and vertices, and the whole nonpositive region without assuming beforehand that an optimum must lie on the chosen edge. Existence also follows directly from compactness of $\mathcal{P}$ and continuity of η. □

9 Reproducible rational arithmetic (L2 and L7)

The included `check_width_optimum.py` uses only Python's standard library integers and `Fraction`. It starts from the rational intervals below, proves that they contain the named exact constants, and does not import a saved numerical result or an optimizer. To make the arithmetic reproducible by an independent implementation, we specify every enclosure formula.

9.1 Trigonometric gates

Use the intervals

$$\begin{aligned}
T_- &= 0.739085133215160641655312087673873404013 4,\\
T_+ &= 0.739085133215160641655312087673873404013 5,\\
q_- &= 0.195020091350606930079807125913415101936 6,\\
q_+ &= 0.195020091350606930079807125913415101936 7,\\
\pi_- &= 3.141592653589793238462643383279502884197 1,\\
\pi_+ &= 3.141592653589793238462643383279502884197 2.
\end{aligned}$$

Let

$$C_m(x) = \sum_{j=0}^{m-1} \frac{(-1)^j x^{2j}}{(2j)!}, \quad S_m(x) = \sum_{j=0}^{m-1} \frac{(-1)^j x^{2j+1}}{(2j+1)!}, \quad A_m(x) = \sum_{j=0}^{m-1} \frac{(-1)^j x^{2j+1}}{2j+1}.$$

For $0 < x < 1$ the terms decrease to zero, so an even number of terms is a lower bound and an odd number is an upper bound. Exact rational comparisons give

$$C_{30}(T_-) > T_-, \qquad C_{31}(T_+) < T_+.$$

Monotonicity of $\cos t - t$ isolates τ. Sine is increasing on $(0,1)$, so $s_- = S_{30}(T_-) < \sin\tau < S_{31}(T_+) = s_+$. The decreasing map $s \mapsto (1-s)/(1+s)$ and exact comparisons give

$$q_- < \frac{1-s_+}{1+s_+} < q < \frac{1-s_-}{1+s_-} < q_+.$$

In particular $q_- > 3/17$ and $q_+ < 1/5$ are checked as rational inequalities.

For π use Machin's identity $\pi = 16\arctan(1/5) - 4\arctan(1/239)$. Here is a branch justification: two tangent doublings give $\tan(4\arctan(1/5)) = 120/119$, and the tangent of $4\arctan(1/5) - \arctan(1/239)$ is exactly one. This angle lies strictly between zero and $4/5 < \pi/2$ (the elementary inequality $\pi > 2$ follows, for example, by comparing a semicircle with its diameter). Positivity follows from $\arctan(1/5) > 1/5 - (1/5)^3/3$ and $\arctan(1/239) < 1/239$. The angle is therefore $\pi/4$. Alternating sums give the exact gates

$$\pi_- < 16A_{32}(1/5) - 4A_{31}(1/239) < \pi < 16A_{31}(1/5) - 4A_{32}(1/239) < \pi_+.$$

9.2 Integral certificates

For each β_i use $s_0 = 1 + q_-$, $v = (1-q_-)/s_0$, $w = (s_0 - 2\beta_i)/s_0$. These satisfy $0 < w < v < 1$. The convergent binomial expansion is

$$\sqrt{1-y^2} = 1 - \sum_{j\ge 1} c_j y^{2j}, \qquad c_j = \frac{\binom{2j}{j}}{4^j(2j-1)} > 0.$$

Its coefficients decrease, since $c_{j+1}/c_j = (2j-1)/(2j+2) < 1$. Termwise integration is valid uniformly on $[w,v]$, which lies inside the radius of convergence. Define the rational quantities

$$\begin{aligned}
P &= v - w - \sum_{j=1}^{80} c_j \frac{v^{2j+1} - w^{2j+1}}{2j+1},\\
T &= (v-w)c_{81}\frac{v^{162}}{1-v^2}, \qquad I_- = \frac{s_0^2}{4}(P-T), \quad I_+ = \frac{s_0^2}{4}P.
\end{aligned}$$

For every $y \in [w, v]$ the omitted positive tail is at most $c_{81} v^{162}/(1 - v^2)$, hence

$$I_- \le \int_{q_-}^{\beta_i} \sqrt{x(1 + q_- - x)}\, \mathrm{d}x \le I_+.$$

Write $D(r, \beta) = \int_r^\beta D_r(x)\, \mathrm{d}x$. The endpoint and integrand bounds are $|D_r(r)| \le 2$ and $|\partial_r D_r(x)| \le 2$. Differentiation under the integral therefore gives $|\partial_r D(r, \beta)| \le 4$ on $r \in [q_-, q_+]$. Thus valid rational endpoints for D_i are

$$d_{i,-} = s_0(\beta_i - q_-) - (\beta_i^2 - q_-^2)/2 - 2I_+ - 4(q_+ - q_-),$$
$$d_{i,+} = s_0(\beta_i - q_-) - (\beta_i^2 - q_-^2)/2 - 2I_- + 4(q_+ - q_-).$$

Before further propagation the script enlarges these endpoints to 30-place rationals: for an interval $[l, r]$ it uses

$$\left[\frac{\lfloor 10^{30} l \rfloor - 1}{10^{30}}, \frac{\lceil 10^{30} r \rceil + 1}{10^{30}}\right].$$

The extra unit makes the printed enclosures strict; it is an outward rounding, never a decimal approximation used as an exact input value.

9.3 Polynomial signs, multipliers and final enclosure

Interval addition and subtraction use matching and opposite endpoints; multiplication takes the minimum and maximum of all four products. Division by a positive interval multiplies by its reversed reciprocal. Positive powers use monotonic endpoints. These operations remain safe when a variable appears repeatedly, although dependency can widen them. Evaluate G at $0, a, z_-, z_+$ using the intervals for D_i. Their signs are respectively $+, -, +, -$ with strict rational margins. With $Z = [z_-, z_+]$ use the interval vector

$$U = ([a - z_+, a - z_-],\ Z,\ [b - z_+, b - z_-]).$$

Evaluate $F(U)$, $\rho = F(U)/\mathcal{L}(U)$ and $D_i - 24U_i^2 - 432\rho$ outwardly. These certify $F(u) > 0$ and the two positive multipliers in Lemma 8.1. The accompanying `certificate.json` gives the rational endpoints of these gates, including the four signs, so they can be checked without interpreting floating output. The enclosed quantities include

$$0.00136820131190299900276 < D_1 < 0.00136820131190299900279,$$
$$0.00196196296152142894680 < D_2 < 0.00196196296152142894683,$$
$$0.00241410289623904895464 < D_3 < 0.00241410289623904895467,$$
$$0.00000038803240421408705 < \eta_{\mathrm{width}} < 0.00000038803240421408841.$$

Finally enclose $C_{\mathrm{term}} = \tau(1 + q)/(2\pi)$ by multiplying the positive intervals for $\tau, 1 + q$ and dividing by opposite endpoints of 2π. Add the interval for ρ/π. Exact comparisons with the two rationals in (4) prove the strict enclosure in that equation. The script `verify.py` checks those literal target endpoints, the certificate content and the input/output manifest; a replay that merely prints a plausible decimal is not the acceptance condition.

10 The ordered passage to the limit (L8)

For any fixed positive width vector with strict macroscopic margins $\delta_i = \beta_{i+1} - \beta_i - h_i - h_{i+1} > 0$, choose an integer N such that $N \ge 102$, $N(\beta_1 - q) > 1$, and $N\delta_i \ge 1$. For every $n \ge N$,

$$\ell_1/n > \beta_1 - 1/n > q, \qquad \frac{\ell_{i+1} - \ell_i}{n} > \beta_{i+1} - \beta_i - 1/n \ge h_i + h_{i+1}.$$

Also $n - \ell_3 \geq 77n/100 \geq 2$. The gates (18) hold for every such integer n. Proposition 7.1 therefore implies, at fixed h,

$$\liminf_{n\to\infty} \frac{M_{n,4}}{n^2} \geq C_{\mathrm{term}} + \eta(h), \qquad \liminf_{n\to\infty} \frac{R^*(n)}{n^2} \geq C_{\mathrm{term}} + \eta(h). \tag{25}$$

The optimum u saturates both macroscopic constraints and cannot be substituted blindly into the finite gates. For any fixed $0 < t < 1$, use $h = tu$. Its strict margins are $(1-t)a$ and $(1-t)b$. The rational gates $\beta_1 - q > a$ and $1/a > 102$ show that

$$N(t) = \left\lceil \frac{1}{(1-t)a} \right\rceil$$

is a sufficient threshold. Apply (25) with this fixed t and only then let $t \uparrow 1$. By continuity $\eta(tu) \to \eta(u)$, so

$$\liminf_{n} M_{n,4}/n^2 \geq C_{\mathrm{term}} + \eta_{\mathrm{width}}, \qquad \liminf_{n} R^*(n)/n^2 \geq C_{\mathrm{term}} + \eta_{\mathrm{width}}. \tag{26}$$

The quantity $M_{n,4}$ depends on the three cutoffs and common order, not on the auxiliary widths. The same liminf is bounded for every fixed t; no minimum over orders has been exchanged with a limit.

For completeness, unchanged u fails both finite separation gates at every $n = 50000j + 1$, $j \geq 1$. The exact floors are

$$(\ell_1, \ell_2, \ell_3) = (10455j, 10907j, 11500j),$$

and their normalized differences are respectively $a(n-1)/n < a$ and $b(n-1)/n < b$. This explains why the order of limits is essential. It is not a failure of (26).

11 Existence of the global limit

We include only the line comparison and concatenation argument needed to identify the liminf in (26) with C_*. No finite-word linear programming characterization or upper-bound construction is needed.

For a finite list of nonnegative marks $v_1, \ldots, v_N$, let $b(v)$ be the least span of line points carrying those marks, with

$$|x_i - x_j| \geq \sqrt{v_i v_j} \quad (i \neq j). \tag{27}$$

Equal marks still label distinct objects; zero marks may coincide. For a fixed order w, the minimal increasing placement is

$$x_1 = 0, \qquad x_i = \max_{j<i} \{x_j + \sqrt{w_i w_j}\}.$$

It satisfies all constraints. Induction shows that every feasible placement in this order starting at zero has coordinates at least these. Thus the minimum span for a fixed order is attained; minimizing over finitely many orders proves attainment of $b(v)$. Deleting or decreasing marks cannot increase b; scaling marks by $h \geq 0$ scales b by h. For $h > 0$ the latter follows by scaling positions in both directions, and for $h = 0$ both sides are zero. These assertions also hold for marks temporarily larger than one.

For marks in $[0, 1]$, let $B(v)$ be the infimum circumference of a circle on which both directed arcs between each pair are at least $\sqrt{v_i v_j}$. Cutting a feasible circle gives a feasible line placement. Conversely close an optimal line placement with a gap of length one: one arc is the line distance and the complementary arc contains the new gap, at least $1 \geq \sqrt{v_i v_j}$. Hence

$$b(v) \leq B(v) \leq b(v) + 1. \tag{28}$$

Write $b_n = b(1/n, 2/n, \dots, 1)$. From a full feasible original circle at radius R, scale angles by $(R+n)/(2n)$. The lower kernel bound in Lemma 2.1 produces a B-circle of circumference $\pi(R+n)/n$. Conversely a B-circle of circumference $b_n + 1$ exists by the preceding gap construction. Scale it to angular circumference 2π and set $R = n(b_n + 1)/\pi$. Every directed arc is then at least $2\sqrt{uv}/R$, hence at least $\theta_R(u, v)$ by the upper kernel bound. The cosine law gives full original feasibility for all pairs. Taking infima proves for every $n \geq 3$

$$\frac{b_n}{\pi n} - \frac{1}{n} \leq \frac{R^*(n)}{n^2} \leq \frac{b_n + 1}{\pi n}. \tag{29}$$

Lemma 11.1. *The limit $E_\infty = \lim_n b_n/n$ exists and equals $\inf_{n\geq 2}(b_n + 1)/n$. Consequently $C_* = E_\infty/\pi$ exists.*

Proof. Fix $n \geq 2$. For any $N \geq n$, put $k = \lceil N/n \rceil$ and concatenate k copies of an optimal b_n arrangement with inter-block gaps one. All cross-block distances are at least one, so all pair constraints hold. Its span is $k(b_n+1)-1$, with exactly k copies of each of the n marks. Retain the N smallest marks. In sorted mark order their ith value is $\lceil i/k \rceil/n$. Scale all positions and marks by $h = nk/N$, so this value is at least i/N. Decrease the retained marks to i/N, assigning each label once (ties may be broken arbitrarily). Feasibility is preserved. Thus

$$b_N \leq \frac{nk}{N}\,[k(b_n + 1) - 1].$$

Since $k/N \to 1/n$, division by N gives $\limsup_N b_N/N \leq (b_n + 1)/n$ for every fixed n. Equally spaced points with gaps one give $0 \leq b_N/N \leq 1$. Choose n along an unbounded subsequence attaining the liminf; then $\limsup_N b_N/N \leq \liminf_n b_n/n$, proving existence. The preceding upper bounds and $n \to \infty$ identify the limit with the displayed infimum. Equation (29) now proves the final assertion for the full geometric problem. □

Combining Lemma 11.1, (26) and the rational enclosure proves Theorem 1.1. The concatenation construction uses actual labeled points and every integer N, not only a subsequence or a coupling with correct marginals.

12 Verification scope and provenance

The proof establishes exact theorems and corollaries. The parameter enclosures are finite rational computations supporting their stated signs and constants. Bounded tour and measure checks test implementation invariants and failure modes; they do not prove the universal statements by enumeration. Neither endpoint of a geometric asymptotic interval is claimed sharp. The optimized width value is sharp only for the function η on the specified polyhedron at the fixed cutoffs.

The analytic provenance is the immutable scientific snapshot

`3beb8d70c5b3748d370a92855847bdf574e5a14f`.

The four source notes are identified in the bibliography. The published target is

`paper_assets/asymptotic_sequel/ringmin_asymptotic.tex`,

Proposition 7; the older finite-paper v2 source is not used as the target. The package includes a dependency map, `DEPENDENCIES.md`, recording source sections, hypotheses and verification limits. Its `MANIFEST.json` records input and output SHA-256 hashes, and `VERIFICATION.md` reports only checks actually run. These supporting files are included alongside this source and PDF; no checkout history or old task dossier is required.

References

[1] M. Falconi, Ringmin asymptotic sequel, arXiv:2609.13630, Proposition 7.

[2] M. Falconi, *Optimized induced-terminal-subset asymptotic lower bound*, Sections 5–7, scientific snapshot above; `research/INDUCED_SUBSET_ASYMPTOTIC_LOWER_BOUND.md`.

[3] M. Falconi, *Quantitative stability for one common pair of terminal chain tours*, Sections 2–5, 11.1–11.3, same snapshot; `research/COMMON_CHAIN_QUANTITATIVE_STABILITY.md`.

[4] M. Falconi, *A three-level common-chain lower bound from two shared cutoff budgets*, Sections 7, 9, 11–12, same snapshot; `research/THREE_LEVEL_COMMON_CHAIN.md`.

[5] M. Falconi, *The global asymptotic limit and finite-word variational certificates*, Sections 1–3, same snapshot; `research/GLOBAL_ASYMPTOTIC_VARIATIONAL_LIMIT.md`.